CHILAKALA SURYA PRAKASH

Java Backend Developer | Spring Boot · Microservices · AWS

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PROFESSIONAL SUMMARY

Motivated Java Backend Developer with hands-on experience building production-grade microservices using **Spring Boot, Apache Kafka, Redis, and AWS EC2**. Proven ability to architect event-driven financial systems with ACID transactional integrity, integrate **Google Gemini AI** for intelligent natural language processing, and enforce stateless security via **JWT and Spring Security** — all backed by an automated **GitHub Actions CI/CD pipeline** and full cloud deployment. Strong foundation in **OOP, Java Collections, Streams API**, RESTful API design, and test-driven development with JUnit 5 and Mockito.

TECHNICAL SKILLS

Programming Languages:	Java (17/21), Python, SQL — OOP & Design Principles, Java Collections Framework, Streams & Lambdas, Exception Handling
Backend & Frameworks:	Spring Boot 3.x, Spring Security 6, Spring Data JPA / Hibernate 6, RESTful API Design, Microservices Architecture
Messaging & Caching:	Apache Kafka (Event Streaming), Redis (Idempotency & Distributed Locking), Asynchronous Processing
Databases:	PostgreSQL, MySQL, H2 (In-Memory), ACID Transactions, Custom JPQL & Aggregation Queries
Security & AI Integration:	JWT (HMAC-SHA256), BCrypt Encryption, Google Gemini 2.5 Flash API, LLM Prompt Engineering, Spring Security Filter Chains
DevOps, Cloud & Testing:	Docker, Docker Compose, GitHub Actions (CI/CD), AWS (EC2, VPC, IAM, EBS), JUnit 5, Mockito, Maven, Postman

WORK EXPERIENCE

Data Analytics & Visualization Simulation — Accenture North America (Forage) May 2024 – Jul 2024

Remote

- Completed a job simulation advising a hypothetical social media client, functioning as a Data Analyst within Accenture's analytics practice.
- Cleaned, modeled, and analyzed 7 datasets to uncover content trends, applying data preparation and EDA techniques to surface strategic business insights.
- Prepared and delivered a PowerPoint deck and video presentation communicating findings clearly to client and internal stakeholders, demonstrating business-facing data storytelling skills.

PROJECTS

Digital Wallet Microservice · Java 17 · Spring Boot · Apache Kafka · Redis · PostgreSQL · Docker GitHub ↗

- Architected a containerized, event-driven ledger microservice in **Java 17** and **Spring Boot 3.2** handling peer-to-peer fund transfers with strict **ACID transactional integrity** and `BigDecimal` monetary precision.
- Engineered network-level idempotency protection using a custom servlet filter and **Redis atomic locks (SETNX, 24h TTL)**, intercepting retries and preventing duplicate debits/double-spending with HTTP 409 responses.
- Built an asynchronous event streaming pipeline broadcasting **PII-masked audit events (`****1001`)** to **Apache Kafka** (`transaction-events`), safely decoupling ledger persistence from external consumers.
- Automated the CI/CD pipeline via **GitHub Actions** to run unit, slice (`@WebMvcTest`, `@DataJpaTest`), and H2 integration tests alongside multi-stage Docker builds on every commit; deployed the 4-container stack to **AWS EC2**, configuring **VPC Security Groups, IAM**, and 4GB EBS Linux swap memory to optimize JVM and Kafka broker performance.

AI-Powered Expense Tracker & Budget Guardrail API · Java 21 · Spring Boot · Google Gemini AI · JWT GitHub ↗

- Developed a personal finance REST microservice in **Java 21** and **Spring Boot 3.4**, integrating **Google Gemini 2.5 Flash** via zero-shot prompt engineering to autonomously convert natural language text into strongly-typed JPA entities.
- Engineered a real-time budget guardrail engine evaluating cumulative month-to-date spending on every transaction write, dispatching proactive threshold alerts (`SAFE`, `WARNING`, `EXCEEDED`) to prevent overspending.
- Implemented stateless authentication and authorization using custom **Spring Security filter chains**, **HMAC-SHA256 signed JWT tokens**, and **BCrypt encryption** for secure multi-tenant data access.
- Optimized multi-tenant financial reporting using database-level **JPQL aggregations (`COALESCE(SUM())`, `GROUP BY`)**, offloading heavy calculations from application memory and reducing API response latency.

Mental Health Risk Prediction Pipeline · Python · scikit-learn · XGBoost · K-Means GitHub ↗

- Engineered an end-to-end ML classification pipeline on DASS-42 psychological survey data, computing Depression, Anxiety, and Stress sub-scores normalized with age and **10 TIPI personality traits**.
- Applied unsupervised **K-Means clustering** (optimized via Elbow & Silhouette analysis, *k=3*) to categorize respondents into 3 clinical severity tiers (*Mild, Moderate, Severe*), deriving ground-truth target labels for supervised training.
- Trained and benchmarked 4 algorithms (**XGBoost, Random Forest, MLP, Logistic Regression**) via `RandomizedSearchCV` and 5-fold cross-validation; **XGBoost** achieved top performance with **97–99% accuracy and weighted F1-score**.

EDUCATION

Bachelor of Engineering — Computer Science & Engineering July 2025

Gopalan College of Engineering and Management, Bangalore **CGPA: 7.1 / 10**

CERTIFICATIONS

- 100 Days of Code: The Complete Python Pro Bootcamp — Udemy
- Data Visualization with Power BI — Great Learning Academy
- Data Analysis with Python — freeCodeCamp.org
- AWS Certified Developer - Associate — Infosys Springboard